Sahaj Bhadja

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Summary

Software engineer with 2+ years building scalable **enterprise AI platforms and full-stack systems** using Python, TypeScript, React, Kubernetes, Docker, PostgreSQL, and Redis across backend APIs, cloud infrastructure, integrations, and frontend features.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*

- Architected a multi-tenant enterprise AI platform using Python, FastAPI, TypeScript, PostgreSQL, and cloud storage, supporting 500+ users while reducing dashboard latency from 1.2 seconds to 350 milliseconds.
- Built fault-tolerant backend services processing 1,000+ AI jobs and 5,000+ monthly integrations with explicit state transitions, validation, failure isolation, and CI/CD-backed releases achieving 99%+ reliability.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*

- Developed a full-stack AI application with a React frontend and modular Python APIs, serving 150+ instructors and handling 10,000+ daily requests with stable overall production performance.
- Designed concurrent batch-processing infrastructure and switchable model integrations, evaluating 10,000+ submissions while reducing feedback turnaround 60% and maintaining 90%+ consistency through automated testing and deployment workflows.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 – May 2025*

- Built large-scale scalable distributed model-training and simulation pipelines processing 100,000+ trajectory logs, improving long-horizon planning 18% through Python, data processing, algorithm evaluation, and rigorous reproducible experimentation.
- Designed a spatial-temporal transformer architecture integrating map and agent context, validating 10,000 scenarios and improving closed-loop stability 15% through rigorous systematic model testing and performance analysis.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*

- Engineered scalable cross-platform microservices and cloud integrations using object-oriented patterns, AWS Glue, and S3, reducing response times 30%, technical debt 20%, and data-processing overhead across distributed systems.
- Automated 20+ comprehensive unit and integration tests across Linux, Windows, and macOS through global CI/CD pipelines, collaborating with Microsoft and GitHub engineers while improving processing efficiency 68%.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*

- Implemented eight modular backend services and React features for authentication, resource allocation, billing, permissions, and notifications, synchronizing two million records across locations through REST APIs and WebSockets.
- Containerized applications with Docker and automated testing and reliable deployment through Jenkins, achieving 95%+ build validation while reducing deployment time 50% and code-review turnaround 25% across releases.

Skills

Languages: Python, TypeScript, JavaScript, Java, C++, SQL, Shell, Bash
Front End: React.js, Next.js, HTML, CSS, TailwindCSS, responsive design, componentization
Backend: FastAPI, Flask, Node.js, Express.js, REST APIs, microservices, distributed systems, networking
Data: PostgreSQL, Redis, MongoDB, MySQL, SQL Server, schema design, query optimization, data pipelines
Platform: Kubernetes, Docker, AWS, Azure, GCP, CI/CD, Git, Linux, operating systems, algorithms, data structures, Agile

Projects

Chat2SQL Code | *Java, TypeScript, PostgreSQL, Redis, Docker*

- Extended a unified AI-powered development platform supporting PostgreSQL, Redis, and 10+ databases, implementing Java backend and TypeScript frontend features, schema exploration, query validation, and Docker-based cross-platform deployment.

FinNexus Live | *TypeScript, Kubernetes, MongoDB*

- Built a scalable TypeScript platform using Kubernetes and MongoDB, supporting 50,000+ users and one million transactions through secure APIs, real-time synchronization, automated deployment, and responsive React features.

CaptionCraft Live | *Python, PyTorch, Model Evaluation*

- Built and evaluated an AI application across four models, 10,000 images, and 100,000 captions, developing Python inference APIs and systematic testing workflows that improved measured accuracy from 81% to 96%.